SYSTEM HIGH-LEVEL DESIGN (HLD)

Brain-Controlled Autonomous Wheelchair (BCI-Wheelchair)

A Real-Time Motor Imagery Brain-Computer Interface Integrated with a 3D Cyberbotics Webots Digital Twin

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Abstract

This document defines the production-grade High-Level Design (HLD) for the Brain-Controlled Autonomous Wheelchair project. The system translates non-invasive sensorimotor electroencephalography (EEG) signals corresponding to mental movement imagery (Left Hand, Right Hand, Both Feet, Idle Rest) into deterministic navigation commands. These commands drive a real-time, physics-based 3D digital twin wheelchair inside the Cyberbotics Webots simulation framework equipped with active collision-avoidance autonomy. This specification details the five foundational subsystems, mathematical formulations, inter-process communication protocols, interface contracts, safety safeguards, and team delivery milestones.

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1 Executive Overview & Problem Formulation

1.1 Clinical & Societal Motivation

Individuals afflicted with severe neurodegenerative disorders (e.g., Amyotrophic Lateral Sclerosis - ALS, severe cerebral palsy) or spinal cord injuries (quadriplegia/tetraplegia) lose motor pathways from the motor cortex to peripheral muscles. Traditional assistive devices such as joysticks, sip-and-puff straws, and chin controls require residual physical control that advanced-stage patients do not possess.

Brain-Computer Interfaces (BCIs) provide a direct communication and control channel between the central nervous system and external robotic hardware, bypassing damaged peripheral neuromuscular pathways entirely.

1.2 Project Scope & Objectives

The primary objective of this project is to develop, validate, and demonstrate an end-to-end, deterministic, low-latency assistive wheelchair system powered by Motor Imagery (MI) EEG signals and integrated with a physical digital twin in Webots.

- **In-Scope:**
  1. Ingestion and standardization of 64-channel international 10–10 system EDF+ EEG records from the PhysioNet EEGMMIDB dataset.
  2. Bandpass filtering, channel selection (sensorimotor strip), and event epoch extraction.
  3. Classical spatial filtering (Common Spatial Patterns – CSP) with Linear Discriminant Analysis (LDA) and Support Vector Machines (SVM).
  4. Deep learning architecture implementation using compact Convolutional Neural Networks (PyTorch EEGNet).
  5. Real-time sliding window circular buffer simulating continuous EEG streaming.
  6. Low-latency Inter-Process Communication (IPC) utilizing ZeroMQ/TCP.
  7. A 3D digital twin of a differential-drive wheelchair in Cyberbotics Webots featuring indoor obstacle scenarios and autonomous safety overrides.
- **Out-of-Scope:** Invasive intra-cortical electrode arrays (ECoG/Utah arrays), hardware fabrication of physical chair chassis (addressed via verified Webots physical simulation), and multi-day continuous adaptive online training.

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2 System Requirements Specification

2.1 Functional Requirements (FR)

FR-1 Signal Ingestion: The system shall ingest multi-channel EEG recordings in standard European Data Format (EDF+), sampling at 160 Hz across 64 channels.

FR-2 Temporal & Spatial Filtering: The ingestion pipeline shall isolate sensorimotor μ (8–12 Hz) and β (16–24 Hz) rhythms using a zero-phase 4th-order Butterworth bandpass filter (8.0–30.0 Hz) and extract central electrodes ($C_3, C_z, C_4, FC_3, FC_4, CP_3, CP_4$).

FR-3 Motor Imagery Decoding: The system shall classify at least four mental states:

- State 1: Left Hand Motor Imagery → Turn Left (Rotate CCW).

- State 2: Right Hand Motor Imagery → Turn Right (Rotate CW).
- State 3: Both Feet / Fists Imagery → Move Forward (Linear Drive).
- State 4: Rest / Baseline State → Neutral / Hold (Active Braking).

FR-4 Streaming Buffer: The real-time engine shall process EEG streams using a sliding circular window of length $T_{window} = 2.0$ s updated every $T_{stride} = 250$ ms (4 Hz decision rate).

FR-5 Command Smoothing: A temporal debounce state machine shall suppress transient classification noise by requiring $N = 2$ consecutive identical intent outputs before issuing a motor transition.

FR-6 IPC Bridge: The classified navigation intent and confidence metrics shall be transmitted over ZeroMQ (PUB/SUB) or TCP socket using a standardized JSON schema.

FR-7 Kinematic Execution: The Webots robot controller shall parse incoming JSON commands and calculate differential wheel angular velocities (ω_L, ω_R) based on physical wheel radius R and track width L.

FR-8 Active Safety Override: The digital twin shall continuously sample onboard distance sensors (LiDAR/Ultrasonic). If an obstacle is detected within critical distance ($d_{thresh} \leq 0.35$ m), the autonomous safety system shall override BCI commands and bring the chair to an immediate halt.

2.2 Non-Functional Requirements (NFR)

NFR-1 End-to-End Latency: The total latency from window ingestion to motor actuation must not exceed 400 ms (Target: < 280 ms).

NFR-2 Classification Accuracy: Subject-specific 5-fold cross-validation accuracy on binary classification (Left vs. Right) must achieve $\geq 80\%$, with multi-class accuracy $\geq 70\%$.

NFR-3 Fail-Safe Operation: If no IPC command is received within 1000 ms (heartbeat timeout), the wheelchair must transition into a fail-safe STOP state.

NFR-4 Modularity: All 5 subsystems must maintain decoupled APIs, enabling independent unit testing, benchmarking, and replacement.

NFR-5 Deterministic Resource Utilization: The inference engine must execute comfortably on commodity CPU/GPU hardware without exceeding 1.5 GB RAM footprint.

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3 High-Level System Architecture

3.1 Architectural Block Diagram

The architecture follows a decoupled, publisher-subscriber pipeline spanning five modular subsystems:

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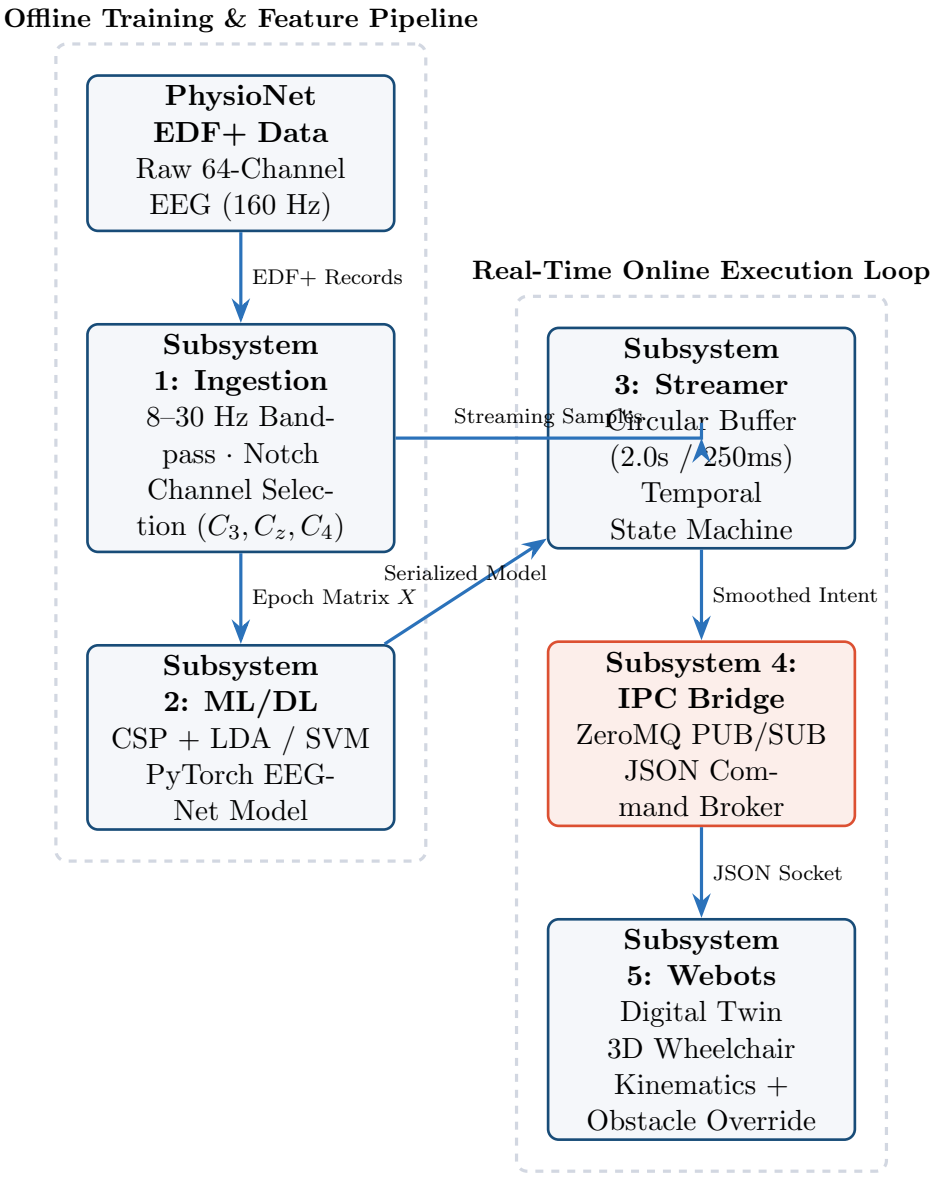


Figure 1: End-to-End High-Level System Architecture and Inter-Subsystem Data Flow.

4 Neurophysiological Foundation & Mathematical Formulations

4.1 Sensorimotor Rhythms: ERD and ERS

The neurophysiological foundation relies on **Event-Related Desynchronization (ERD)** and **Event-Related Synchronization (ERS)**. In the resting state, pyramidal neurons in the primary motor cortex fire synchronously, generating distinct oscillatory rhythms: μ band (8–12 Hz) and β band (16–24 Hz).

When a subject mentally imagines limb movement (e.g., right hand):

- **Contralateral Motor Cortex (C_3 - Left Hemisphere):** Neural populations desynchronize, causing a marked attenuation of power (**ERD**).
- **Ipsilateral Motor Cortex (C_4 - Right Hemisphere):** Neural populations exhibit idle or increased synchronization, resulting in power maintenance or increase (**ERS**).

4.2 Common Spatial Patterns (CSP) Formulation

Common Spatial Patterns (CSP) maximizes the variance of bandpass-filtered signals for one class while simultaneously minimizing it for the other class.

Let $\mathbf{X}_1 \in \mathbb{R}^{C\times S}$ and $\mathbf{X}_2 \in \mathbb{R}^{C\times S}$ represent single-trial EEG matrices for two motor imagery classes (e.g., Left vs. Right), where C is the number of channels and S is samples per epoch. The normalized spatial covariance matrix for class $c \in \{1, 2\}$ is:

$$\mathbf{\Sigma}_c = \frac{\mathbf{X}_c\mathbf{X}_c^T}{\mathrm{Tr}(\mathbf{X}_c\mathbf{X}_c^T)} \tag{1}$$

The composite covariance across classes is:

$$\mathbf{\Sigma} = \bar{\mathbf{\Sigma}}_1 + \bar{\mathbf{\Sigma}}_2 = \mathbf{U}_0\mathbf{\Lambda}\mathbf{U}_0^T \tag{2}$$

where $\mathbf{U}_0$ is the matrix of eigenvectors and $\mathbf{\Lambda}$ is the diagonal matrix of eigenvalues sorted in descending order. The whitening transformation matrix is defined as:

$$\mathbf{P} = \mathbf{\Lambda}^{-1/2}\mathbf{U}_0^T \tag{3}$$

Transforming the individual covariance matrices:

$$\mathbf{S}_1 = \mathbf{P}\bar{\mathbf{\Sigma}}_1\mathbf{P}^T, \quad \mathbf{S}_2 = \mathbf{P}\bar{\mathbf{\Sigma}}_2\mathbf{P}^T \tag{4}$$

Matrices $\mathbf{S}_1$ and $\mathbf{S}_2$ share common eigenvectors $\mathbf{B}$:

$$\mathbf{S}_1 = \mathbf{B}\mathbf{\Lambda}_1\mathbf{B}^T, \quad \mathbf{S}_2 = \mathbf{B}\mathbf{\Lambda}_2\mathbf{B}^T \quad \text{where} \quad \mathbf{\Lambda}_1 + \mathbf{\Lambda}_2 = \mathbf{I} \tag{5}$$

The projection matrix $\mathbf{W} \in \mathbb{R}^{2m\times C}$ is formed by taking the first m and last m columns of $\mathbf{B}^T\mathbf{P}$. The feature vector $\mathbf{f}_p \in \mathbb{R}^{2m}$ for an epoch $\mathbf{X}$ is extracted via normalized log-variance:

$$\mathbf{f}_p = \log\left(\frac{\mathrm{var}(\mathbf{W}_p\mathbf{X})}{\sum_{i=1}^{2m}\mathrm{var}(\mathbf{W}_i\mathbf{X})}\right), \quad p = 1, \ldots, 2m \tag{6}$$

4.3 Differential Drive Kinematics (Webots Wheelchair)

The 3D virtual wheelchair in Cyberbotics Webots is modeled as a non-holonomic differential-drive mobile base characterized by:

- R: Radius of driving wheels ($R = 0.15$ m).

- L: Axle track width between wheels ($L = 0.55$ m).
- v: Linear forward velocity ($v \in [0.0, 0.8]$ m/s).
- ω: Angular rotational velocity ($\omega \in [-1.2, 1.2]$ rad/s).

The left and right wheel angular velocities (ω_L, ω_R) in radians per second are computed as:

$$\omega_L = \frac{v - \frac{\omega \cdot L}{2}}{R}, \qquad \omega_R = \frac{v + \frac{\omega \cdot L}{2}}{R} \tag{7}$$

Decoded Brain Intent	Wheelchair Action	Kinematic Setting	Actuator Response
Left Hand Imagery (T1)	Turn Left	$v = 0.0$ m/s, $\omega = +0.8$ rad/s	$\omega_L = -1.47$ rad/s, $\omega_R = +1.47$ ra
Right Hand Imagery (T2)	Turn Right	$v = 0.0$ m/s, $\omega = -0.8$ rad/s	$\omega_L = +1.47$ rad/s, $\omega_R = -1.47$ ra
Both Feet / Fists (T1/T2)	Move Forward	$v = 0.4$ m/s, $\omega = 0.0$ rad/s	$\omega_L = +2.67$ rad/s, $\omega_R = +2.67$ ra
Rest / Idle State (T0)	Neutral / Stop	$v = 0.0$ m/s, $\omega = 0.0$ rad/s	$\omega_L = 0.00$ rad/s, $\omega_R = 0.00$ rad/s

Table 1: Intent-to-Kinematic Actuation Mapping Matrix.

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5 Subsystem Detailed Architecture

5.1 Subsystem 1: Offline Ingestion & Preprocessing

- **Input:** Raw EDF+ recordings from PhysioNet EEGMMIDB (Runs 04, 08, 12 for Left/Right; Runs 06, 10, 14 for Feet/Fists).
- **Filtering Pipeline:**
  1. Zero-phase IIR Butterworth bandpass filter: $f_{low} = 8.0$ Hz, $f_{high} = 30.0$ Hz, Order = 4.
  2. Notch filter: $f_{notch} = 50.0$ Hz/60.0 Hz to eliminate powerline interference.
- **Electrode Subsetting:** Extraction of 7 sensorimotor channels: $[C_3, C_z, C_4, FC_3, FC_4, CP_3, CP_4]$.
- **Epoching Window:** Time window $[t_{min} = 1.0\text{ s}, t_{max} = 3.5\text{ s}]$ post visual cue to isolate steady-state motor imagery, producing 3D tensor $\mathbf{X} \in \mathbb{R}^{N_{trials} \times N_{channels} \times N_{samples}}$.

5.2 Subsystem 2: Machine Learning & Deep Learning Engine

The project benchmark incorporates two complementary modeling paradigms:

1. **Classical Pipeline (CSP + Classifier):**
   - Spatial projection via $2m = 4$ CSP spatial filters.
   - Classification via Linear Discriminant Analysis (LDA) with automatic shrinkage (Ledoit-Wolf lemma) or Support Vector Classifier (RBF kernel, $C = 1.0$).
2. **Deep Learning Pipeline (PyTorch EEGNet):**
   - **Block 1 (Temporal Convolution):** Kernel $(1, 32)$ learning bandpass frequency filters.
   - **Block 2 (Depthwise Spatial Convolution):** Kernel $(C, 1)$ learning spatial channel combinations per temporal filter (depth multiplier $D = 2$).

- **Block 3 (Separable Convolution):** Depthwise $(1, 16)$ followed by pointwise $(1, 1)$, batch normalization, Exponential Linear Units (ELU), and spatial dropout ($p = 0.25$).
- **Dense Softmax Layer:** Produces posterior probabilities $\mathbf{p} \in \mathbb{R}^K$ across navigation classes.

5.3 Subsystem 3: Real-Time Streaming & Sliding Window Engine

To simulate an active EEG headset in production:

- **Circular Buffer Ring:** A ring buffer stores incoming samples $\mathbf{S} \in \mathbb{R}^{C \times N_{buffer}}$, configured with capacity $N_{buffer} = 320$ samples (2.0 s at 160 Hz).
- **Sliding Stride:** Every 40 samples (250 ms), the latest 320 samples are passed to the band-pass filter and model inference engine.
- **Temporal State Machine (Debouncing):**

$$C_{final} = \begin{cases} \hat{y}_t & \text{if } \hat{y}_t = \hat{y}_{t-1} \text{ and } P(\hat{y}_t) \geq \tau_{confidence} \\ C_{previous} & \text{otherwise} \end{cases} \tag{8}$$

  where $\tau_{confidence} = 0.65$. This prevents high-frequency motor chattering.

5.4 Subsystem 4: IPC / Network Communication Bridge

The interface between the Python ML environment and Webots is decoupled via a high-speed socket bridge:

- **Transport Protocol:** ZeroMQ PUB/SUB socket or standard TCP loopback (‘127.0.0.1:5555‘).
- **Serialization Format:** UTF-8 JSON packet.
- **Publishing Rate:** 4 Hz (synchronized with sliding window stride).

```
{
  "timestamp": 1729331000.254,
  "sequence_id": 1420,
  "intent": "TURN_LEFT",
  "confidence": 0.884,
  "probabilities": {
    "REST": 0.041,
    "TURN_LEFT": 0.884,
    "TURN_RIGHT": 0.052,
    "FORWARD": 0.023
  },
  "safety_flag": false
}
```

Listing 1: Standardized JSON IPC Command Payload.

5.5 Subsystem 5: Cyberbotics Webots Digital Twin

- **Environment Arena:** An indoor apartment or clinic environment featuring corridors, doorways, rooms, and furniture obstacles.
- **Wheelchair Node:** Differential-drive chassis containing:

  - Two active rotational motors (‘left$_w$heel$_m$otor‘, ‘right$_w$heel$_m$otor‘)invelocitycontrolmode.

  - Two passive caster wheels for pitch/roll stability.
  - Distance sensor array: Front-left, Front-center, Front-right sensors.

- **Active Braking Override Logic:** If $\min(d_{front_left}, d_{front_center}, d_{front_right}) < 0.35$ m, the controller sets $\omega_L = 0, \omega_R = 0$ regardless of incoming BCI forward commands, preventing collisions during user distraction.

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6 Timing Analysis & Latency Budget

To guarantee seamless user experience and safe robotic mobility, the end-to-end latency budget is strictly budgeted as detailed in Table 2.

Pipeline Stage	Component Description	Allocated Latency	Worst-Case Latency
Stage 1	Circular Buffer Stride Advance	250.0 ms	250.0 ms
Stage 2	Temporal Bandpass Filtering (IIR/FIR)	4.0 ms	10.0 ms
Stage 3	Feature Extraction (CSP Projection)	1.5 ms	4.0 ms
Stage 4	Model Inference (LDA / PyTorch Forward)	2.5 ms	8.0 ms
Stage 5	Debounce Logic & State Machine	0.5 ms	1.0 ms
Stage 6	IPC Serialization & TCP Transmission	1.5 ms	5.0 ms
Stage 7	Webots Motor Actuator Step	16.0 ms	32.0 ms
Total	**Closed-Loop System Latency**	**276.0 ms**	**310.0 ms**

Table 2: End-to-End System Latency Budget Breakdown.

Safety Margin: The total latency of ≈ 276 ms comfortably satisfies the target threshold of < 400 ms, ensuring real-time responsiveness without perceived lag.

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7 Team Work Breakdown & Implementation Roadmap

7.1 Standardized Repository Directory Layout

The repository structure enforces separation of concerns across subsystems:

```
Brain-Controlled-Wheelchair/
|-- DATASET/                    <- Core benchmark subjects (S001-S010)
|-- docs/                       <- Architecture, HLD, and LLD specifications
|-- src/
|   |-- ingestion/              <- Subsystem 1: EDF readers, MNE filters, epoching
|   |   |-- __init__.py
|   |   |-- edf_loader.py
|   |   +-- preprocessor.py
|   |-- models/                 <- Subsystem 2: CSP, LDA, SVM, and PyTorch EEGNet
|   |   |-- __init__.py
|   |   |-- csp_classifier.py
|   |   |-- eegnet.py
|   |   +-- trainer.py
```

```
|   |-- streaming/              <- Subsystem 3: Sliding buffer & state machine
|   |   |-- __init__.py
|   |   |-- ring_buffer.py
|   |   +-- inference_engine.py
|   |-- ipc/                    <- Subsystem 4: ZeroMQ / Socket command broker
|   |   |-- __init__.py
|   |   |-- zmq_publisher.py
|   |   +-- command_schema.py
|   +-- webots_controller/      <- Subsystem 5: Webots robot driver
|       |-- wheelchair_driver.py
|       +-- safety_override.py
|-- simulation/                 <- Webots world files, 3D assets, robot protos
|   |-- worlds/
|   |   +-- indoor_arena.wbt
|   +-- protos/
|       +-- Wheelchair.proto
|-- tests/                      <- Unit tests, synthetic stream tests
|-- scripts/                    <- Automated dataset downloaders and benchmarks
|   +-- download_dataset.py
|-- MASTER_PROJECT_PLAN.md
|-- requirements.txt
+-- README.md
```

7.2 Teammate Work Breakdown Matrix

Subsystem / Role	Core Engineering Responsibilities	Key Deliverables
Engineer 1: Biosignals & Preprocessing	Ingestion pipeline, MNE-Python integration, Butterworth/Notch filtering, epoching, PSD analysis.	‘edf$_{l}oader.py$‘, ‘$preprocessor.py$‘, $WelchPSDplot$
Engineer 2: ML/DL Models	CSP implementation, LDA/SVM tuning, PyTorch EEGNet training, cross-validation metrics.	‘csp$_{c}lassifier.py$‘, ‘$eegnet.py$‘, $serializedmodelw$
Engineer 3: Streaming & IPC Bridge	Circular buffer implementation, temporal state machine, ZeroMQ socket publisher.	‘ring$_{b}uffer.py$‘, ‘$zmq_{p}ublisher.py$‘, $latencyprofil$
Engineer 4: Webots Simulation	3D indoor environment, wheelchair kinematics, distance sensor active braking override.	‘Wheelchair.proto‘, ‘indoor$_{a}rena.wbt$‘, ‘$wheelchair_{d}river.py$‘.

Table 3: Engineering Team Roles and Work Allocation.

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8 Risk Management & Failure Mode Analysis (FMEA)

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Failure Mode	Root Cause	Severity	Mitigation Strategy
High-Frequency Jitter	Marginal classification probabilities near decision boundary.	Moderate	2-step temporal debounce filter and confidence threshold (≥ 0.65).
Socket Disconnection	Network interrupt or crashed inference loop.	Critical	Webots watchdog timer: automatic emergency brake if no heartbeat within 1.0s.
Obstacle Collision	BCI misclassification or user inattention near walls.	Critical	Autonomous sensor override takes hardware priority over user intent.
Subject Variability	Non-stationary EEG cross-subject drift.	Moderate	Subject-specific CSP calibration runs prior to navigation sessions.

Table 4: Failure Mode and Effects Analysis (FMEA).

9 Sign-Off & Document Approvals

Role	Name / Signature	Date
Lead Systems Architect	_____	_____
ML / Neuro-Engineering Lead	_____	_____
Robotics & Simulation Lead	_____	_____
Quality Assurance Lead	_____	_____